

☰ CS50 Manual Pages

Manual pages for the C standard library, the C POSIX library, and the CS50 Library for those less comfortable.

☐ frequently used in CS50

aio.h

[aio_cancel](#) - cancel an outstanding asynchronous I/O request
[aio_error](#) - get error status of asynchronous I/O operation
[aio_fsync](#) - asynchronous file synchronization
[aio_init](#) - asynchronous I/O initialization
[aio_read](#) - asynchronous read
[aio_return](#) - get return status of asynchronous I/O operation
[aio_suspend](#) - wait for asynchronous I/O operation or timeout
[aio_write](#) - asynchronous write
[lio_listio](#) - initiate a list of I/O requests

arpa/inet.h

[byteorder](#) - convert values between host and network byte order
[htonl](#) - convert values between host and network byte order
[htons](#) - convert values between host and network byte order
[inet_net_ntop](#) - Internet network number conversion
[inet_net_pton](#) - Internet network number conversion
[inet_ntop](#) - convert IPv4 and IPv6 addresses from binary to text form
[inet_pton](#) - convert IPv4 and IPv6 addresses from text to binary form
[ntohl](#) - convert values between host and network byte order
[ntohs](#) - convert values between host and network byte order

assert.h

`_Static_assert` - fail compilation if assertion is false
`assert` - abort the program if assertion is false
`assert_perror` - test errnum and abort
`static_assert` - fail compilation if assertion is false

complex.h

`cabs` - absolute value of a complex number
`cabsf` - absolute value of a complex number
`cabsl` - absolute value of a complex number
`cacos` - complex arc cosine
`cacosf` - complex arc cosine
`cacosh` - complex arc hyperbolic cosine
`cacoshf` - complex arc hyperbolic cosine
`cacoshl` - complex arc hyperbolic cosine
`cacosl` - complex arc cosine
`carg` - calculate the complex argument
`cargf` - calculate the complex argument
`cargl` - calculate the complex argument
`casin` - complex arc sine
`casinf` - complex arc sine
`casinh` - complex arc sine hyperbolic
`casinhf` - complex arc sine hyperbolic
`casinhl` - complex arc sine hyperbolic
`casinl` - complex arc sine
`catan` - complex arc tangents
`catanf` - complex arc tangents
`catanh` - complex arc tangents hyperbolic
`catanhf` - complex arc tangents hyperbolic
`catanhl` - complex arc tangents hyperbolic
`catanl` - complex arc tangents
`ccos` - complex cosine function
`ccosf` - complex cosine function
`ccosh` - complex hyperbolic cosine
`ccoshf` - complex hyperbolic cosine
`ccoshl` - complex hyperbolic cosine
`ccosl` - complex cosine function
`cexp` - complex exponential function
`cexp2` - base-2 exponent of a complex number

`cexp2f` - base-2 exponent of a complex number
`cexp2l` - base-2 exponent of a complex number
`cexpf` - complex exponential function
`cexpl` - complex exponential function
`cimag` - get imaginary part of a complex number
`cimagf` - get imaginary part of a complex number
`cimagl` - get imaginary part of a complex number
`clog` - natural logarithm of a complex number
`clog10` - base-10 logarithm of a complex number
`clog10f` - base-10 logarithm of a complex number
`clog10l` - base-10 logarithm of a complex number
`clog2` - base-2 logarithm of a complex number
`clog2f` - base-2 logarithm of a complex number
`clog2l` - base-2 logarithm of a complex number
`clogf` - natural logarithm of a complex number
`clogl` - natural logarithm of a complex number
`complex` - basics of complex mathematics
`conj` - calculate the complex conjugate
`conjf` - calculate the complex conjugate
`conjl` - calculate the complex conjugate
`cpow` - complex power function
`cpowf` - complex power function
`cpowl` - complex power function
`cproj` - project into Riemann Sphere
`cprojf` - project into Riemann Sphere
`cprojl` - project into Riemann Sphere
`creal` - get real part of a complex number
`crealf` - get real part of a complex number
`creall` - get real part of a complex number
`csin` - complex sine function
`csinf` - complex sine function
`csinh` - complex hyperbolic sine
`csinhf` - complex hyperbolic sine
`csinhl` - complex hyperbolic sine
`csinl` - complex sine function
`csqrt` - complex square root
`csqrtf` - complex square root
`csqrtl` - complex square root
`ctan` - complex tangent function
`ctanf` - complex tangent function

`ctanh` - complex hyperbolic tangent
`ctanhf` - complex hyperbolic tangent
`ctanhl` - complex hyperbolic tangent
`ctanl` - complex tangent function

cs50.h

`get_char` - prompts user for a line of text from stdin and returns...
`get_double` - prompts user for a line of text from stdin and returns...
`get_float` - prompts user for a line of text from stdin and returns...
`get_int` - prompts user for a line of text from stdin and returns...
`get_long` - prompts user for a line of text from stdin and returns...
`get_long_long` - prompts user for a line of text from stdin and returns...
`get_string` - prompts user for a line of text from stdin and returns...

ctype.h

`isalnum` - character classification functions
`isalnum_l` - character classification functions
`isalpha` - character classification functions
`isalpha_l` - character classification functions
`isascii` - character classification functions
`isascii_l` - character classification functions
`isblank` - character classification functions
`isblank_l` - character classification functions
`iscntrl` - character classification functions
`iscntrl_l` - character classification functions
`isdigit` - character classification functions
`isdigit_l` - character classification functions
`isgraph` - character classification functions
`isgraph_l` - character classification functions
`islower` - character classification functions
`islower_l` - character classification functions
`isprint` - character classification functions
`isprint_l` - character classification functions
`ispunct` - character classification functions
`ispunct_l` - character classification functions

`isspace` - character classification functions
`isspace_l` - character classification functions
`isupper` - character classification functions
`isupper_l` - character classification functions
`isxdigit` - character classification functions
`isxdigit_l` - character classification functions
`toascii` - convert character to ASCII
`tolower` - convert uppercase or lowercase
`tolower_l` - convert uppercase or lowercase
`toupper` - convert uppercase or lowercase
`toupper_l` - convert uppercase or lowercase

`dirent.h`

`alphasort` - scan a directory for matching entries
`getdirentries` - get directory entries in a filesystem-independent format
`readdir` - read a directory
`readdir_r` - read a directory
`scandir` - scan a directory for matching entries
`scandirat` - scan a directory for matching entries
`seekdir` - set the position of the next `readdir()` call in the `dir...`
`telldir` - return current location in directory stream
`versionsort` - scan a directory for matching entries

`dlfcn.h`

`dladdr` - translate address to symbolic information
`dladdr1` - translate address to symbolic information
`dlclose` - open and close a shared object
`dlerror` - obtain error diagnostic for functions in the `dlopen` API
`dlmopen` - open and close a shared object
`dlopen` - open and close a shared object
`dlsym` - obtain address of a symbol in a shared object or execu...
`dlvsym` - obtain address of a symbol in a shared object or execu...

`errno.h`

`errno` - number of last error

`program_invocation_name` - obtain name used to invoke calling program

`program_invocation_short_name` - obtain name used to invoke calling program

`fcntl.h`

`arm_fadvise` - predeclare an access pattern for file data

`arm_fadvise64_64` - predeclare an access pattern for file data

`arm_sync_file_range` - sync a file segment with disk

`creat` - open and possibly create a file

`fadvise64` - predeclare an access pattern for file data

`fadvise64_64` - predeclare an access pattern for file data

`fallocate` - manipulate file space

`fcntl` - manipulate file descriptor

`fcntl64` - manipulate file descriptor

`name_to_handle_at` - obtain handle for a pathname and open file via a handle

`open` - open and possibly create a file

`open_by_handle_at` - obtain handle for a pathname and open file via a handle

`openat` - open and possibly create a file

`posix_fadvise` - predeclare an access pattern for file data

`posix_fallocate` - allocate file space

`readahead` - initiate file readahead into page cache

`splice` - splice data to/from a pipe

`sync_file_range` - sync a file segment with disk

`sync_file_range2` - sync a file segment with disk

`tee` - duplicating pipe content

`vmsplice` - splice user pages to/from a pipe

`fenv.h`

`feclearexcept` - floating-point rounding and exception handling

`fedisableexcept` - floating-point rounding and exception handling

`feenableexcept` - floating-point rounding and exception handling

`fegetenv` - floating-point rounding and exception handling

`fegetexcept` - floating-point rounding and exception handling

`fegetexceptflag` - floating-point rounding and exception handling
`fegetround` - floating-point rounding and exception handling
`feholdexcept` - floating-point rounding and exception handling
`fenv` - floating-point rounding and exception handling
`feraiseexcept` - floating-point rounding and exception handling
`fesetenv` - floating-point rounding and exception handling
`fesetexceptflag` - floating-point rounding and exception handling
`fesetround` - floating-point rounding and exception handling
`fetestexcept` - floating-point rounding and exception handling
`feupdateenv` - floating-point rounding and exception handling

`fmtmsg.h`

`addseverity` - introduce new severity classes
`fmtmsg` - print formatted error messages

`fnmatch.h`

`fnmatch` - match filename or pathname

`ftw.h`

`ftw` - file tree walk
`nftw` - file tree walk

`glob.h`

`glob` - find pathnames matching a pattern, free memory from gl...
`globfree` - find pathnames matching a pattern, free memory from gl...

`grp.h`

`fgetgrent_r` - get group file entry reentrantly
`getgrent_r` - get group file entry reentrantly
`getgrouplist` - get list of groups to which a user belongs

`putgrent` - write a group database entry to a file

iconv.h

`iconv` - perform character set conversion

`iconv_close` - deallocate descriptor for character set conversion

`iconv_open` - allocate descriptor for character set conversion

inttypes.h

`strtoimax` - convert string to integer

`strtoumax` - convert string to integer

langinfo.h

`nl_langinfo` - query language and locale information

`nl_langinfo_l` - query language and locale information

libgen.h

`basename` - parse pathname components

`dirname` - parse pathname components

limits.h

`MB_LEN_MAX` - maximum multibyte length of a character across all loc...

`realpath` - return the canonicalized absolute pathname

locale.h

`duplocale` - duplicate a locale object

`freelocale` - create, modify, and free a locale object

`locale` - description of multilanguage support

`localeconv` - get numeric formatting information

- `newlocale` - create, modify, and free a locale object
- `setlocale` - set the current locale
- `uselocale` - set/get the locale for the calling thread

math.h

```
acos - arc cosine function
acosf - arc cosine function
acosh - inverse hyperbolic cosine function
acoshf - inverse hyperbolic cosine function
acoshl - inverse hyperbolic cosine function
acosl - arc cosine function
asin - arc sine function
asinf - arc sine function
asinh - inverse hyperbolic sine function
asinhf - inverse hyperbolic sine function
asinhf - inverse hyperbolic sine function
asinhf - inverse hyperbolic sine function
asinhf - inverse hyperbolic sine function
asinhf - inverse hyperbolic sine function
atan - arc tangent function
atan2 - arc tangent function of two variables
atan2f - arc tangent function of two variables
atan2l - arc tangent function of two variables
atanf - arc tangent function
atanh - inverse hyperbolic tangent function
atanhf - inverse hyperbolic tangent function
atanhl - inverse hyperbolic tangent function
atanl - arc tangent function
cbrt - cube root function
cbrtf - cube root function
cbrtl - cube root function
ceil - ceiling function: smallest integral value not less tha...
ceilf - ceiling function: smallest integral value not less tha...
ceill - ceiling function: smallest integral value not less tha...
copysign - copy sign of a number
copysignf - copy sign of a number
copysignl - copy sign of a number
cos - cosine function
cosf - cosine function
cosh - hyperbolic cosine function
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`coshf` - hyperbolic cosine function
`coshl` - hyperbolic cosine function
`cosl` - cosine function
`drem` - floating-point remainder function
`dremf` - floating-point remainder function
`dreml` - floating-point remainder function
`erf` - error function
`erfc` - complementary error function
`erfcf` - complementary error function
`erfcl` - complementary error function
`erff` - error function
`erfl` - error function
`exp` - base-e exponential function
`exp10` - base-10 exponential function
`exp10f` - base-10 exponential function
`exp10l` - base-10 exponential function
`exp2` - base-2 exponential function
`exp2f` - base-2 exponential function
`exp2l` - base-2 exponential function
`expf` - base-e exponential function
`expl` - base-e exponential function
`expm1` - exponential minus 1
`expm1f` - exponential minus 1
`expm1l` - exponential minus 1
`fabs` - absolute value of floating-point number
`fabsf` - absolute value of floating-point number
`fabsl` - absolute value of floating-point number
`fdim` - positive difference
`fdimf` - positive difference
`fdiml` - positive difference
`finite` - BSD floating-point classification functions
`finitef` - BSD floating-point classification functions
`finitel` - BSD floating-point classification functions
`floor` - largest integral value not greater than argument
`floorf` - largest integral value not greater than argument
`floorl` - largest integral value not greater than argument
`fma` - floating-point multiply and add
`fmaf` - floating-point multiply and add
`fmal` - floating-point multiply and add
`fmax` - determine maximum of two floating-point numbers

`fmaxf` - determine maximum of two floating-point numbers
`fmaxl` - determine maximum of two floating-point numbers
`fmin` - determine minimum of two floating-point numbers
`fminf` - determine minimum of two floating-point numbers
`fminl` - determine minimum of two floating-point numbers
`fmod` - floating-point remainder function
`fmodf` - floating-point remainder function
`fmodl` - floating-point remainder function
`fpclassify` - floating-point classification macros
`frexp` - convert floating-point number to fractional and integr...
`frexpf` - convert floating-point number to fractional and integr...
`frexpl` - convert floating-point number to fractional and integr...
`gamma` - (logarithm of the) gamma function
`gammaf` - (logarithm of the) gamma function
`gammal` - (logarithm of the) gamma function
`HUGE_VAL` - floating-point constants
`HUGE_VALF` - floating-point constants
`HUGE_VALL` - floating-point constants
`hypot` - Euclidean distance function
`hypotf` - Euclidean distance function
`hypotl` - Euclidean distance function
`ilogb` - get integer exponent of a floating-point value
`ilogbf` - get integer exponent of a floating-point value
`ilogbl` - get integer exponent of a floating-point value
`INFINITY` - floating-point constants
`isfinite` - floating-point classification macros
`isgreater` - floating-point relational tests without exception for NaN
`isgreaterequal` - floating-point relational tests without exception for NaN
NaN
`isinf` - floating-point classification macros
`isinff` - BSD floating-point classification functions
`isinfl` - BSD floating-point classification functions
`isless` - floating-point relational tests without exception for NaN
`islessequal` - floating-point relational tests without exception for NaN
`islessgreater` - floating-point relational tests without exception for NaN
NaN
`isnan` - floating-point classification macros
`isnanf` - BSD floating-point classification functions
`isnanl` - BSD floating-point classification functions
`isnormal` - floating-point classification macros

`isunordered` - floating-point relational tests without exception for NaN

`j0` - Bessel functions of the first kind

`j0f` - Bessel functions of the first kind

`j0l` - Bessel functions of the first kind

`j1` - Bessel functions of the first kind

`j1f` - Bessel functions of the first kind

`j1l` - Bessel functions of the first kind

`jn` - Bessel functions of the first kind

`jnf` - Bessel functions of the first kind

`jnl` - Bessel functions of the first kind

`ldexp` - multiply floating-point number by integral power of 2

`ldexpf` - multiply floating-point number by integral power of 2

`ldexpl` - multiply floating-point number by integral power of 2

`lgamma` - log gamma function

`lgamma_r` - log gamma function

`lgammaf` - log gamma function

`lgammaf_r` - log gamma function

`lgammal` - log gamma function

`lgammal_r` - log gamma function

`llrint` - round to nearest integer

`llrintf` - round to nearest integer

`llrintl` - round to nearest integer

`llround` - round to nearest integer

`llroundf` - round to nearest integer

`llroundl` - round to nearest integer

`log` - natural logarithmic function

`log10` - base-10 logarithmic function

`log10f` - base-10 logarithmic function

`log10l` - base-10 logarithmic function

`log1p` - logarithm of 1 plus argument

`log1pf` - logarithm of 1 plus argument

`log1pl` - logarithm of 1 plus argument

`log2` - base-2 logarithmic function

`log2f` - base-2 logarithmic function

`log2l` - base-2 logarithmic function

`logb` - get exponent of a floating-point value

`logbf` - get exponent of a floating-point value

`logbl` - get exponent of a floating-point value

`logf` - natural logarithmic function

`logl` - natural logarithmic function

`lrint` - round to nearest integer
`lrintf` - round to nearest integer
`lrintl` - round to nearest integer
`lround` - round to nearest integer
`lroundf` - round to nearest integer
`lroundl` - round to nearest integer
`math_error` - detecting errors from mathematical functions
`matherr` - SVID math library exception handling
`modf` - extract signed integral and fractional values from flo...
`modff` - extract signed integral and fractional values from flo...
`modfl` - extract signed integral and fractional values from flo...
`NAN` - floating-point constants
`nan` - return 'Not a Number'
`nanf` - return 'Not a Number'
`nanl` - return 'Not a Number'
`nearbyint` - round to nearest integer
`nearbyintf` - round to nearest integer
`nearbyintl` - round to nearest integer
`nextafter` - floating-point number manipulation
`nextafterf` - floating-point number manipulation
`nextafterl` - floating-point number manipulation
`nextdown` - return next floating-point number toward positive/nega...
`nextdownf` - return next floating-point number toward positive/nega...
`nextdownl` - return next floating-point number toward positive/nega...
`nexttoward` - floating-point number manipulation
`nexttowardf` - floating-point number manipulation
`nexttowardl` - floating-point number manipulation
`nextup` - return next floating-point number toward positive/nega...
`nextupf` - return next floating-point number toward positive/nega...
`nextupl` - return next floating-point number toward positive/nega...
`pow` - power functions
`pow10` - base-10 power functions
`pow10f` - base-10 power functions
`pow10l` - base-10 power functions
`powf` - power functions
`powl` - power functions
`remainder` - floating-point remainder function
`remainderf` - floating-point remainder function
`remainderl` - floating-point remainder function
`remquo` - remainder and part of quotient

`remquof` - remainder and part of quotient
`remquol` - remainder and part of quotient
`rint` - round to nearest integer
`rintf` - round to nearest integer
`rintl` - round to nearest integer
`round` - round to nearest integer, away from zero
`roundf` - round to nearest integer, away from zero
`roundl` - round to nearest integer, away from zero
`scalb` - multiply floating-point number by integral power of ra...
`scalbf` - multiply floating-point number by integral power of ra...
`scalbl` - multiply floating-point number by integral power of ra...
`scalbln` - multiply floating-point number by integral power of radix
`scalblnf` - multiply floating-point number by integral power of radix
`scalblnl` - multiply floating-point number by integral power of radix
`scalbn` - multiply floating-point number by integral power of radix
`scalbnf` - multiply floating-point number by integral power of radix
`scalbnl` - multiply floating-point number by integral power of radix
`signbit` - test sign of a real floating-point number
`signgam` - log gamma function
`significand` - get mantissa of floating-point number
`significandf` - get mantissa of floating-point number
`significandl` - get mantissa of floating-point number
`sin` - sine function
`sincos` - calculate sin and cos simultaneously
`sincosf` - calculate sin and cos simultaneously
`sincosl` - calculate sin and cos simultaneously
`sinf` - sine function
`sinh` - hyperbolic sine function
`sinhf` - hyperbolic sine function
`sinhl` - hyperbolic sine function
`sinl` - sine function
`sqrt` - square root function
`sqrtf` - square root function
`sqrtl` - square root function
`tan` - tangent function
`tanf` - tangent function
`tanh` - hyperbolic tangent function
`tanhf` - hyperbolic tangent function
`tanhl` - hyperbolic tangent function
`tanl` - tangent function

`tgamma` - true gamma function
`tgammaf` - true gamma function
`tgammal` - true gamma function
`trunc` - round to integer, toward zero
`truncf` - round to integer, toward zero
`truncl` - round to integer, toward zero
`y0` - Bessel functions of the second kind
`y0f` - Bessel functions of the second kind
`y0l` - Bessel functions of the second kind
`y1` - Bessel functions of the second kind
`y1f` - Bessel functions of the second kind
`y1l` - Bessel functions of the second kind
`yn` - Bessel functions of the second kind
`ynf` - Bessel functions of the second kind
`ynl` - Bessel functions of the second kind

monetary.h

`strfmon` - convert monetary value to a string
`strfmon_l` - convert monetary value to a string

mqueue.h

`mq_close` - close a message queue descriptor
`mq_getattr` - get/set message queue attributes
`mq_notify` - register for notification when a message is available
`mq_notify` - register for notification when a message is available
`mq_open` - open a message queue
`mq_open` - open a message queue
`mq_receive` - receive a message from a message queue
`mq_send` - send a message to a message queue
`mq_setattr` - get/set message queue attributes
`mq_timedreceive` - receive a message from a message queue
`mq_timedreceive` - receive a message from a message queue
`mq_timedsend` - send a message to a message queue
`mq_timedsend` - send a message to a message queue
`mq_unlink` - remove a message queue
`mq_unlink` - remove a message queue

net/if.h

`if_freenameindex` - get network interface names and indexes

`if_indextoname` - mappings between network interface names and indexes

`if_nameindex` - get network interface names and indexes

`if_nametoindex` - mappings between network interface names and indexes

netdb.h

`endhostent` - get network host entry

`endnetent` - get network entry

`endnetgrent` - handle network group entries

`endprotoent` - get protocol entry

`endrpcent` - get RPC entry

`endservent` - get service entry

`gai_cancel` - asynchronous network address and service translation

`gai_error` - asynchronous network address and service translation

`gai_suspend` - asynchronous network address and service translation

`getaddrinfo_a` - asynchronous network address and service translation

`gethostbyaddr` - get network host entry

`gethostbyaddr_r` - get network host entry

`gethostbyname` - get network host entry

`gethostbyname2` - get network host entry

`gethostbyname2_r` - get network host entry

`gethostbyname_r` - get network host entry

`gethostent` - get network host entry

`gethostent_r` - get network host entry

`getnetbyaddr` - get network entry

`getnetbyaddr_r` - get network entry (reentrant)

`getnetbyname` - get network entry

`getnetbyname_r` - get network entry (reentrant)

`getnetent` - get network entry

`getnetent_r` - get network entry (reentrant)

`getnetgrent` - handle network group entries

`getnetgrent_r` - handle network group entries

`getprotobyname` - get protocol entry

`getprotobyname_r` - get protocol entry (reentrant)

`getprotobynumber` - get protocol entry

`getprotobynumber_r` - get protocol entry (reentrant)
`getprotoent` - get protocol entry
`getprotoent_r` - get protocol entry (reentrant)
`getrpcbyname` - get RPC entry
`getrpcbyname_r` - get RPC entry (reentrant)
`getrpcbynumber` - get RPC entry
`getrpcbynumber_r` - get RPC entry (reentrant)
`getrpccent` - get RPC entry
`getrpccent_r` - get RPC entry (reentrant)
`getservbyname` - get service entry
`getservbyname_r` - get service entry (reentrant)
`getservbyport` - get service entry
`getservbyport_r` - get service entry (reentrant)
`getservent` - get service entry
`getservent_r` - get service entry (reentrant)
`h_errno` - get network host entry
`herror` - get network host entry
`hstrerror` - get network host entry
`innetgr` - handle network group entries
`rexec` - return stream to a remote command
`rexec_af` - return stream to a remote command
`sethostent` - get network host entry
`setnetent` - get network entry
`setnetgrent` - handle network group entries
`setprotoent` - get protocol entry
`setrpccent` - get RPC entry
`setservent` - get service entry

netinet/in.h

`dn_comp` - resolver routines
`dn_expand` - resolver routines
`res_init` - resolver routines
`res_mkquery` - resolver routines
`res_nclose` - resolver routines
`res_ninit` - resolver routines
`res_nmkquery` - resolver routines
`res_nquery` - resolver routines
`res_nquerydomain` - resolver routines

[res_nsearch](#) - resolver routines
[res_nsend](#) - resolver routines
[res_query](#) - resolver routines
[res_querydomain](#) - resolver routines
[res_search](#) - resolver routines
[res_send](#) - resolver routines
[resolver](#) - resolver routines

nl_types.h

[catclose](#) - open/close a message catalog
[catgets](#) - get message from a message catalog
[catopen](#) - open/close a message catalog

poll.h

[poll](#) - wait for some event on a file descriptor
[ppoll](#) - wait for some event on a file descriptor

pthread.h

[pthread_attr_destroy](#) - initialize and destroy thread attributes object
[pthread_attr_getaffinity_np](#) - set/get CPU affinity attribute in thread
at...
[pthread_attr_getdetachstate](#) - set/get detach state attribute in thread
at...
[pthread_attr_getguardsize](#) - set/get guard size attribute in thread
attrib...
[pthread_attr_getinheritsched](#) - set/get inherit-scheduler attribute in
thr...
[pthread_attr_getschedparam](#) - set/get scheduling parameter attributes in
t...
[pthread_attr_getschedpolicy](#) - set/get scheduling policy attribute in
thre...
[pthread_attr_getscope](#) - set/get contention scope attribute in thread
attr...

`pthread_attr_getsigmask_np` - set/get signal mask attribute in thread attr...

`pthread_attr_getstack` - set/get stack attributes in thread attributes object

`pthread_attr_getstackaddr` - set/get stack address attribute in thread att...

`pthread_attr_getstacksize` - set/get stack size attribute in thread attrib...

`pthread_attr_init` - initialize and destroy thread attributes object

`pthread_attr_setaffinity_np` - set/get CPU affinity attribute in thread at...

`pthread_attr_setdetachstate` - set/get detach state attribute in thread at...

`pthread_attr_setguardsize` - set/get guard size attribute in thread attrib...

`pthread_attr_setinheritsched` - set/get inherit-scheduler attribute in thr...

`pthread_attr_setschedparam` - set/get scheduling parameter attributes in t...

`pthread_attr_setschedpolicy` - set/get scheduling policy attribute in thre...

`pthread_attr_setscope` - set/get contention scope attribute in thread attr...

`pthread_attr_setsigmask_np` - set/get signal mask attribute in thread attr...

`pthread_attr_setstack` - set/get stack attributes in thread attributes object

`pthread_attr_setstackaddr` - set/get stack address attribute in thread att...

`pthread_attr_setstacksize` - set/get stack size attribute in thread attrib...

`pthread_cancel` - send a cancelation request to a thread

`pthread_cleanup_pop` - push and pop thread cancelation clean-up handlers

`pthread_cleanup_pop_restore_np` - push and pop thread cancelation clean-up...

`pthread_cleanup_push` - push and pop thread cancelation clean-up handlers

`pthread_cleanup_push_defer_np` - push and pop thread cancelation clean-up ...

`pthread_create` - create a new thread

`pthread_detach` - detach a thread

`pthread_equal` - compare thread IDs

`pthread_exit` - terminate calling thread

`pthread_getaffinity_np` - set/get CPU affinity of a thread

`pthread_getattr_default_np` - get or set default thread-creation attributes

`pthread_getattr_np` - get attributes of created thread

`pthread_getconcurrency` - set/get the concurrency level

`pthread_getcpuclockid` - retrieve ID of a thread's CPU time clock

`pthread_getname_np` - set/get the name of a thread

`pthread_getschedparam` - set/get scheduling policy and parameters of a thread

`pthread_join` - join with a terminated thread

`pthread_kill_other_threads_np` - terminate all other threads in process

`pthread_mutex_consistent` - make a robust mutex consistent

`pthread_mutex_consistent_np` - make a robust mutex consistent

`pthread_mutexattr_getpshared` - get/set process-shared mutex attribute

`pthread_mutexattr_getrobust` - get and set the robustness attribute of a m...

`pthread_mutexattr_getrobust_np` - get and set the robustness attribute of ...

`pthread_mutexattr_setpshared` - get/set process-shared mutex attribute

`pthread_mutexattr_setrobust` - get and set the robustness attribute of a m...

`pthread_mutexattr_setrobust_np` - get and set the robustness attribute of ...

`pthread_rwlockattr_getkind_np` - set/get the read-write lock kind of the t...

`pthread_rwlockattr_setkind_np` - set/get the read-write lock kind of the t...

`pthread_self` - obtain ID of the calling thread

`pthread_setaffinity_np` - set/get CPU affinity of a thread

`pthread_setattr_default_np` - get or set default thread-creation attributes

`pthread_setcancelstate` - set cancelability state and type

`pthread_setcanceltype` - set cancelability state and type

`pthread_setconcurrency` - set/get the concurrency level

`pthread_setname_np` - set/get the name of a thread

`pthread_setschedparam` - set/get scheduling policy and parameters of a thread

`pthread_setschedprio` - set scheduling priority of a thread

`pthread_spin_destroy` - initialize or destroy a spin lock
`pthread_spin_init` - initialize or destroy a spin lock
`pthread_spin_lock` - lock and unlock a spin lock
`pthread_spin_trylock` - lock and unlock a spin lock
`pthread_spin_unlock` - lock and unlock a spin lock
`pthread_testcancel` - request delivery of any pending cancelation request
`pthread_timedjoin_np` - try to join with a terminated thread
`pthread_tryjoin_np` - try to join with a terminated thread
`pthread_yield` - yield the processor

pwd.h

`fgetpwent_r` - get passwd file entry reentrantly
`getpwent_r` - get passwd file entry reentrantly

regex.h

`regcomp` - POSIX regex functions
`regerror` - POSIX regex functions
`regex` - POSIX regex functions
`regexec` - POSIX regex functions
`regfree` - POSIX regex functions

sched.h

`__clone2` - create a child process
`clone` - create a child process
`clone2` - create a child process
`clone3` - create a child process
`CPU_ALLOC` - macros for manipulating CPU sets
`CPU_ALLOC_SIZE` - macros for manipulating CPU sets
`CPU_AND` - macros for manipulating CPU sets
`CPU_AND_S` - macros for manipulating CPU sets
`CPU_CLR` - macros for manipulating CPU sets
`CPU_CLR_S` - macros for manipulating CPU sets
`CPU_COUNT` - macros for manipulating CPU sets
`CPU_COUNT_S` - macros for manipulating CPU sets

[CPU_EQUAL](#) - macros for manipulating CPU sets
[CPU_EQUAL_S](#) - macros for manipulating CPU sets
[CPU_FREE](#) - macros for manipulating CPU sets
[CPU_ISSET](#) - macros for manipulating CPU sets
[CPU_ISSET_S](#) - macros for manipulating CPU sets
[CPU_OR](#) - macros for manipulating CPU sets
[CPU_OR_S](#) - macros for manipulating CPU sets
[CPU_SET](#) - macros for manipulating CPU sets
[CPU_SET_S](#) - macros for manipulating CPU sets
[CPU_XOR](#) - macros for manipulating CPU sets
[CPU_XOR_S](#) - macros for manipulating CPU sets
[CPU_ZERO](#) - macros for manipulating CPU sets
[CPU_ZERO_S](#) - macros for manipulating CPU sets
[getcpu](#) - determine CPU and NUMA node on which the calling threa...
[sched_get_priority_max](#) - get static priority range
[sched_get_priority_min](#) - get static priority range
[sched_getaffinity](#) - set and get a thread's CPU affinity mask
[sched_getcpu](#) - determine CPU on which the calling thread is running
[sched_getparam](#) - set and get scheduling parameters
[sched_getscheduler](#) - set and get scheduling policy/parameters
[sched_rr_get_interval](#) - get the SCHED_RR interval for the named process
[sched_setaffinity](#) - set and get a thread's CPU affinity mask
[sched_setparam](#) - set and get scheduling parameters
[sched_setscheduler](#) - set and get scheduling policy/parameters
[sched_yield](#) - yield the processor
[setns](#) - reassociate thread with a namespace
[unshare](#) - disassociate parts of the process execution context

search.h

[hcreate](#) - hash table management
[hcreate_r](#) - hash table management
[hdestroy](#) - hash table management
[hdestroy_r](#) - hash table management
[hsearch](#) - hash table management
[hsearch_r](#) - hash table management
[insque](#) - insert/remove an item from a queue
[lfind](#) - linear search of an array
[lsearch](#) - linear search of an array

`remque` - insert/remove an item from a queue
`tdelete` - manage a binary search tree
`tdestroy` - manage a binary search tree
`tfind` - manage a binary search tree
`tsearch` - manage a binary search tree
`twalk` - manage a binary search tree
`twalk_r` - manage a binary search tree

semaphore.h

`sem_close` - close a named semaphore
`sem_destroy` - destroy an unnamed semaphore
`sem_getvalue` - get the value of a semaphore
`sem_init` - initialize an unnamed semaphore
`sem_open` - initialize and open a named semaphore
`sem_post` - unlock a semaphore
`sem_timedwait` - lock a semaphore
`sem_trywait` - lock a semaphore
`sem_unlink` - remove a named semaphore
`sem_wait` - lock a semaphore

setjmp.h

`longjmp` - performing a nonlocal goto
`setjmp` - performing a nonlocal goto
`siglongjmp` - performing a nonlocal goto
`sigsetjmp` - performing a nonlocal goto

signal.h

`bsd_signal` - signal handling with BSD semantics
`gsignal` - software signal facility
`kill` - send signal to a process
`killpg` - send signal to a process group
`psiginfo` - print signal description
`psignal` - print signal description
`pthread_kill` - send a signal to a thread

`pthread_sigmask` - examine and change mask of blocked signals
`pthread_sigqueue` - queue a signal and data to a thread
`raise` - send a signal to the caller
`rt_sigaction` - examine and change a signal action
`rt_sigpending` - examine pending signals
`rt_sigprocmask` - examine and change blocked signals
`rt_sigsuspend` - wait for a signal
`rt_sigtimedwait` - synchronously wait for queued signals
`sigaction` - examine and change a signal action
`sigaddset` - POSIX signal set operations
`sigaltstack` - set and/or get signal stack context
`sigandset` - POSIX signal set operations
`sigblock` - BSD signal API
`sigdelset` - POSIX signal set operations
`sigemptyset` - POSIX signal set operations
`sigfillset` - POSIX signal set operations
`siggetmask` - BSD signal API
`sighold` - System V signal API
`sigignore` - System V signal API
`siginterrupt` - allow signals to interrupt system calls
`sigisemptyset` - POSIX signal set operations
`sigismember` - POSIX signal set operations
`sigmask` - BSD signal API
`signal` - ANSI C signal handling
`sigorset` - POSIX signal set operations
`sigpause` - atomically release blocked signals and wait for interrupt
`sigpending` - examine pending signals
`sigprocmask` - examine and change blocked signals
`sigqueue` - queue a signal and data to a process
`sigrelse` - System V signal API
`sigset` - System V signal API
`sigsetmask` - BSD signal API
`sigsetops` - POSIX signal set operations
`sigstack` - set and/or get signal stack context
`sigsuspend` - wait for a signal
`sigtimedwait` - synchronously wait for queued signals
`sigvec` - BSD signal API
`sigwait` - wait for a signal
`sigwaitinfo` - synchronously wait for queued signals
`ssignal` - software signal facility

`sysv_signal` - signal handling with System V semantics

spawn.h

`posix_spawn` - spawn a process

`posix_spawnnp` - spawn a process

stdarg.h

`stdarg` - variable argument lists

`va_arg` - variable argument lists

`va_copy` - variable argument lists

`va_end` - variable argument lists

`va_start` - variable argument lists

stddef.h

`offsetof` - offset of a structure member

`wcstoimax` - convert wide-character string to integer

`wcstoumax` - convert wide-character string to integer

stdio.h

`__fbufsize` - interfaces to stdio FILE structure

`__flbf` - interfaces to stdio FILE structure

`__fpending` - interfaces to stdio FILE structure

`__fpurge` - purge a stream

`__freadable` - interfaces to stdio FILE structure

`__freading` - interfaces to stdio FILE structure

`__fsetlocking` - interfaces to stdio FILE structure

`__fwritable` - interfaces to stdio FILE structure

`__fwriting` - interfaces to stdio FILE structure

`_flushlbf` - interfaces to stdio FILE structure

`addmntent` - get filesystem descriptor file entry

`asprintf` - print to allocated string

`clearerr` - check and reset stream status

`clearerr_unlocked` - nonlocking stdio functions
`ctermid` - get controlling terminal name
`dprintf` - formatted output conversion
`endmntent` - get filesystem descriptor file entry
`fclose` - close a stream
`fcloseall` - close all open streams
`fdopen` - stream open functions
`feof` - check and reset stream status
`feof_unlocked` - nonlocking stdio functions
`ferror` - check and reset stream status
`ferror_unlocked` - nonlocking stdio functions
`fflush` - flush a stream
`fflush_unlocked` - nonlocking stdio functions
`fgetc` - input of characters and strings
`fgetc_unlocked` - nonlocking stdio functions
`fgetgrent` - get group file entry
`fgetpos` - reposition a stream
`fgetpwent` - get password file entry
`fgets` - input of characters and strings
`fgets_unlocked` - nonlocking stdio functions
`fgetwc` - read a wide character from a FILE stream
`fgetwc_unlocked` - nonlocking stdio functions
`fgetws_unlocked` - nonlocking stdio functions
`fileno` - obtain file descriptor of a stdio stream
`fileno_unlocked` - nonlocking stdio functions
`flockfile` - lock FILE for stdio
`fmemopen` - open memory as stream
`fopen` - stream open functions
`fopencookie` - open a custom stream
`fprintf` - formatted output conversion
`fpurge` - purge a stream
`fputc` - output of characters and strings
`fputc_unlocked` - nonlocking stdio functions
`fputs` - output of characters and strings
`fputs_unlocked` - nonlocking stdio functions
`fputwc` - write a wide character to a FILE stream
`fputwc_unlocked` - nonlocking stdio functions
`fputws_unlocked` - nonlocking stdio functions
`fread` - binary stream input/output
`fread_unlocked` - nonlocking stdio functions

`freopen` - stream open functions
`fscanf` - input FILE format conversion
`fseek` - reposition a stream
`fseeko` - seek to or report file position
`fsetpos` - reposition a stream
`ftell` - reposition a stream
`ftello` - seek to or report file position
`ftrylockfile` - lock FILE for stdio
`funlockfile` - lock FILE for stdio
`fwprintf` - formatted wide-character output conversion
`fwrite` - binary stream input/output
`fwrite_unlocked` - nonlocking stdio functions
`getc` - input of characters and strings
`getc_unlocked` - nonlocking stdio functions
`getchar` - input of characters and strings
`getchar_unlocked` - nonlocking stdio functions
`getdelim` - delimited string input
`getline` - delimited string input
`getmntent` - get filesystem descriptor file entry
`getmntent_r` - get filesystem descriptor file entry
`gets` - get a string from standard input (DEPRECATED)
`getw` - input and output of words (ints)
`getwc` - read a wide character from a FILE stream
`getwc_unlocked` - nonlocking stdio functions
`getwchar_unlocked` - nonlocking stdio functions
`hasmntopt` - get filesystem descriptor file entry
`open_memstream` - open a dynamic memory buffer stream
`open_wmemstream` - open a dynamic memory buffer stream
`pclose` - pipe stream to or from a process
`perror` - print a system error message
`popen` - pipe stream to or from a process
`printf` - formatted output conversion
`putc` - output of characters and strings
`putc_unlocked` - nonlocking stdio functions
`putchar` - output of characters and strings
`putchar_unlocked` - nonlocking stdio functions
`putpwent` - write a password file entry
`puts` - output of characters and strings
`putw` - input and output of words (ints)
`putwc` - write a wide character to a FILE stream

`putc_unlocked` - nonlocking stdio functions
`putwchar_unlocked` - nonlocking stdio functions
`remove` - remove a file or directory
`rename` - change the name or location of a file
`renameat` - change the name or location of a file
`renameat2` - change the name or location of a file
`rewind` - reposition a stream
`scanf` - input FILE format conversion
`setbuf` - stream buffering operations
`setbuffer` - stream buffering operations
`setlinebuf` - stream buffering operations
`setmntent` - get filesystem descriptor file entry
`setvbuf` - stream buffering operations
`snprintf` - formatted output conversion
`sprintf` - formatted output conversion
`sscanf` - input string format conversion
`stderr` - standard I/O streams
`stdin` - standard I/O streams
`stdio` - standard input/output library functions
`stdio_ext` - interfaces to stdio FILE structure
`stdout` - standard I/O streams
`swprintf` - formatted wide-character output conversion
`sys_errlist` - print a system error message
`sys_nerr` - print a system error message
`tempnam` - create a name for a temporary file
`tmpfile` - create a temporary file
`tmpnam` - create a name for a temporary file
`tmpnam_r` - create a name for a temporary file
`ungetc` - input of characters and strings
`unlock_stdio` - nonlocking stdio functions
`vasprintf` - print to allocated string
`vdprintf` - formatted output conversion
`vfprintf` - formatted output conversion
`vfscanf` - input FILE format conversion
`vfwprintf` - formatted wide-character output conversion
`vprintf` - formatted output conversion
`vscanf` - input FILE format conversion
`vsprintf` - formatted output conversion
`vsprintf` - formatted output conversion
`vsscanf` - input string format conversion

`vswprintf` - formatted wide-character output conversion

`vwprintf` - formatted wide-character output conversion

`wprintf` - formatted wide-character output conversion

`stdlib.h`

`a64l` - convert between long and base-64

`abort` - cause abnormal process termination

`abs` - compute the absolute value of an integer

`aligned_alloc` - allocate aligned memory

`arc4random` - cryptographically-secure pseudorandom number generator

`arc4random_buf` - cryptographically-secure pseudorandom number generator

`arc4random_uniform` - cryptographically-secure pseudorandom number generator

`atexit` - register a function to be called at normal process ter...

`atof` - convert a string to a double

`atoi` - convert a string to an integer

`atol` - convert a string to an integer

`atoll` - convert a string to an integer

`atoq` - convert a string to an integer

`bsearch` - binary search of a sorted array

`calloc` - allocate and free dynamic memory

`canonicalize_file_name` - return the canonicalized absolute pathname

`cfree` - free allocated memory

`clearenv` - clear the environment

`div` - compute quotient and remainder of an integer division

`drand48` - generate uniformly distributed pseudo-random numbers

`drand48_r` - generate uniformly distributed pseudo-random numbers r...

`ecvt` - convert a floating-point number to a string

`ecvt_r` - convert a floating-point number to a string

`erand48` - generate uniformly distributed pseudo-random numbers

`erand48_r` - generate uniformly distributed pseudo-random numbers r...

`exit` - cause normal process termination

`fcvt` - convert a floating-point number to a string

`fcvt_r` - convert a floating-point number to a string

`free` - allocate and free dynamic memory

`gcvt` - convert a floating-point number to a string

`getenv` - get an environment variable

`getloadavg` - get system load averages

`getpt` - open a new pseudoterminal master
`getsubopt` - parse suboption arguments from a string
`grantpt` - grant access to the slave pseudoterminal
`imaxabs` - compute the absolute value of an integer
`imaxdiv` - compute quotient and remainder of an integer division
`initstate` - random number generator
`initstate_r` - reentrant random number generator
`jrand48` - generate uniformly distributed pseudo-random numbers
`jrand48_r` - generate uniformly distributed pseudo-random numbers r...
`l64a` - convert between long and base-64
`labs` - compute the absolute value of an integer
`lcong48` - generate uniformly distributed pseudo-random numbers
`lcong48_r` - generate uniformly distributed pseudo-random numbers r...
`ldiv` - compute quotient and remainder of an integer division
`llabs` - compute the absolute value of an integer
`lldiv` - compute quotient and remainder of an integer division
`lrand48` - generate uniformly distributed pseudo-random numbers
`lrand48_r` - generate uniformly distributed pseudo-random numbers r...
`malloc` - allocate and free dynamic memory
`MB_CUR_MAX` - maximum length of a multibyte character in the current...
`mblen` - determine number of bytes in next multibyte character
`mbstowcs` - convert a multibyte string to a wide-character string
`mbtowc` - convert a multibyte sequence to a wide character
`memalign` - allocate aligned memory
`mkdtemp` - create a unique temporary directory
`mkostemp` - create a unique temporary file
`mkostemps` - create a unique temporary file
`mkstemp` - create a unique temporary file
`mkstemps` - create a unique temporary file
`mktemp` - make a unique temporary filename
`mrand48` - generate uniformly distributed pseudo-random numbers
`mrand48_r` - generate uniformly distributed pseudo-random numbers r...
`nrand48` - generate uniformly distributed pseudo-random numbers
`nrand48_r` - generate uniformly distributed pseudo-random numbers r...
`on_exit` - register a function to be called at normal process ter...
`posix_memalign` - allocate aligned memory
`posix_openpt` - open a pseudoterminal device
`ptsname` - get the name of the slave pseudoterminal
`ptsname_r` - get the name of the slave pseudoterminal
`putenv` - change or add an environment variable

`pvalloc` - allocate aligned memory
`qecvt` - convert a floating-point number to a string
`qecvt_r` - convert a floating-point number to a string
`qfcvt` - convert a floating-point number to a string
`qfcvt_r` - convert a floating-point number to a string
`qgcvt` - convert a floating-point number to a string
`qsort` - sort an array
`qsort_r` - sort an array
`rand` - pseudo-random number generator
`rand_r` - pseudo-random number generator
`random` - random number generator
`random_r` - reentrant random number generator
`realloc` - allocate and free dynamic memory
`reallocarray` - allocate and free dynamic memory
`rpmatch` - determine if the answer to a question is affirmative o...
`secure_getenv` - get an environment variable
`seed48` - generate uniformly distributed pseudo-random numbers
`seed48_r` - generate uniformly distributed pseudo-random numbers r...
`setenv` - change or add an environment variable
`setstate` - random number generator
`setstate_r` - reentrant random number generator
`srand` - pseudo-random number generator
`srand48` - generate uniformly distributed pseudo-random numbers
`srand48_r` - generate uniformly distributed pseudo-random numbers r...
`srandom` - random number generator
`srandom_r` - reentrant random number generator
`strfromd` - convert a floating-point value into a string
`strfromf` - convert a floating-point value into a string
`strfroml` - convert a floating-point value into a string
`strtod` - convert ASCII string to floating-point number
`strtof` - convert ASCII string to floating-point number
`strtoul` - convert a string to a long integer
`strtold` - convert ASCII string to floating-point number
`strtoll` - convert a string to a long integer
`strtouq` - convert a string to a long integer
`strtoul` - convert a string to an unsigned long integer
`strtoull` - convert a string to an unsigned long integer
`strtouq` - convert a string to an unsigned long integer
`system` - execute a shell command
`unlockpt` - unlock a pseudoterminal master/slave pair

`unsetenv` - change or add an environment variable
`valloc` - allocate aligned memory
`wcstombs` - convert a wide-character string to a multibyte string
`wctomb` - convert a wide character to a multibyte sequence

string.h

`bstring` - byte string operations
`memccpy` - copy memory area
`memchr` - scan memory for a character
`memcmp` - compare memory areas
`memcpy` - copy memory area
`memfrob` - frobnicate (obfuscate) a memory area
`memmem` - locate a substring
`memmove` - copy memory area
`mempcpy` - copy memory area
`memrchr` - scan memory for a character
`memset` - fill memory with a constant byte
`rawmemchr` - scan memory for a character
`sigabbrev_np` - return string describing signal
`sigdescr_np` - return string describing signal
`stpcpy` - copy or concatenate a string
`stpncpy` - fill a fixed-size buffer with non-null bytes from a st...
`strcasestr` - locate a substring
`strcat` - copy or concatenate a string
`strchr` - locate character in string
`strchrnul` - locate character in string
`strcmp` - compare two strings
`strcoll` - compare two strings using the current locale
`strcpy` - copy or concatenate a string
`strcspn` - get length of a prefix substring
`strdup` - duplicate a string
`strdupa` - duplicate a string
`strerror` - return string describing error number
`strerror_l` - return string describing error number
`strerror_r` - return string describing error number
`strerrordesc_np` - return string describing error number
`strerrorname_np` - return string describing error number
`strfry` - randomize a string

`strlen` - calculate the length of a string
`strncat` - append non-null bytes from a source array to a string,...
`strncmp` - compare two strings
`strncpy` - fill a fixed-size buffer with non-null bytes from a st...
`strndup` - duplicate a string
`strndupa` - duplicate a string
`strnlen` - determine the length of a fixed-size string
`strpbrk` - search a string for any of a set of bytes
`strrchr` - locate character in string
`strsep` - extract token from string
`strsignal` - return string describing signal
`strspn` - get length of a prefix substring
`strstr` - locate a substring
`strtok` - extract tokens from strings
`strtok_r` - extract tokens from strings
`strverscmp` - compare two version strings
`strxfrm` - string transformation
`sys_siglist` - return string describing signal
`wmempcpy` - copy memory area

strings.h

`bcmp` - compare byte sequences
`bcopy` - copy byte sequence
`bzero` - zero a byte string
`explicit_bzero` - zero a byte string
`ffs` - find first bit set in a word
`ffsl` - find first bit set in a word
`ffsll` - find first bit set in a word
`index` - locate character in string
`rindex` - locate character in string
`strcasecmp` - compare two strings ignoring case
`string` - string operations
`strncasecmp` - compare two strings ignoring case

sys/ipc.h

`ftok` - convert a pathname and a project identifier to a Syste...

sys/mman.h

`madvise` - give advice about use of memory
`memfd_create` - create an anonymous file
`mincore` - determine whether pages are resident in memory
`mlock` - lock and unlock memory
`mlock2` - lock and unlock memory
`mlockall` - lock and unlock memory
`mmap` - map or unmap files or devices into memory
`mmap64` - map or unmap files or devices into memory
`mprotect` - set protection on a region of memory
`mremap` - remap a virtual memory address
`msync` - synchronize a file with a memory map
`munlock` - lock and unlock memory
`munlockall` - lock and unlock memory
`munmap` - map or unmap files or devices into memory
`pkey_alloc` - allocate or free a protection key
`pkey_free` - allocate or free a protection key
`pkey_mprotect` - set protection on a region of memory
`posix_madvise` - give advice about patterns of memory usage
`process_madvise` - give advice about use of memory to a process
`remap_file_pages` - create a nonlinear file mapping
`shm_open` - create/open or unlink POSIX shared memory objects
`shm_unlink` - create/open or unlink POSIX shared memory objects

sys/msg.h

`msgctl` - System V message control operations
`msgget` - get a System V message queue identifier
`msgop` - System V message queue operations
`msgrcv` - System V message queue operations
`msgsnd` - System V message queue operations

sys/resource.h

`getpriority` - get/set program scheduling priority
`getrlimit` - get/set resource limits

`getrusage` - get resource usage
`prlimit` - get/set resource limits
`prlimit64` - get/set resource limits
`setpriority` - get/set program scheduling priority
`setrlimit` - get/set resource limits
`ugetrlimit` - get/set resource limits
`vlimit` - get/set resource limits
`vtimes` - get resource usage

sys/select.h

`_newselect` - synchronous I/O multiplexing
`FD_CLR` - synchronous I/O multiplexing
`FD_ISSET` - synchronous I/O multiplexing
`FD_SET` - synchronous I/O multiplexing
`FD_ZERO` - synchronous I/O multiplexing
`pselect` - synchronous I/O multiplexing
`pselect6` - synchronous I/O multiplexing
`select` - synchronous I/O multiplexing

sys/sem.h

`semctl` - System V semaphore control operations
`semget` - get a System V semaphore set identifier
`semop` - System V semaphore operations
`semtimedop` - System V semaphore operations

sys/shm.h

`shmat` - System V shared memory operations
`shmctl` - System V shared memory control
`shmdt` - System V shared memory operations
`shmget` - allocates a System V shared memory segment
`shmop` - System V shared memory operations

sys/socket.h

`accept` - accept a connection on a socket
`accept4` - accept a connection on a socket
`address_families` - socket address families (domains)
`bind` - bind a name to a socket
`cmsg` - access ancillary data
`CMSG_ALIGN` - access ancillary data
`CMSG_DATA` - access ancillary data
`CMSG_FIRSTHDR` - access ancillary data
`CMSG_LEN` - access ancillary data
`CMSG_NXTHDR` - access ancillary data
`CMSG_SPACE` - access ancillary data
`connect` - initiate a connection on a socket
`ddp` - Linux AppleTalk protocol implementation
`getnameinfo` - address-to-name translation in protocol-independent ma...
`getpeername` - get name of connected peer socket
`getsockname` - get socket name
`getsockopt` - get and set options on sockets
`inet` - Internet address manipulation routines
`inet_addr` - Internet address manipulation routines
`inet_aton` - Internet address manipulation routines
`inet_lnaof` - Internet address manipulation routines
`inet_makeaddr` - Internet address manipulation routines
`inet_netof` - Internet address manipulation routines
`inet_network` - Internet address manipulation routines
`inet_ntoa` - Internet address manipulation routines
`ip` - Linux IPv4 protocol implementation
`ipv6` - Linux IPv6 protocol implementation
`listen` - listen for connections on a socket
`packet` - packet interface on device level
`raw` - Linux IPv4 raw sockets
`recv` - receive a message from a socket
`recvfrom` - receive a message from a socket
`recvmsg` - receive multiple messages on a socket
`recvmsg` - receive a message from a socket
`send` - send a message on a socket
`sendmsg` - send multiple messages on a socket
`sendmsg` - send a message on a socket
`sendto` - send a message on a socket
`setsockopt` - get and set options on sockets
`shutdown` - shut down part of a full-duplex connection

`sock_diag` - obtaining information about sockets
`socketatmark` - determine whether socket is at out-of-band mark
`socket` - create an endpoint for communication
`socket` - Linux socket interface
`socketpair` - create a pair of connected sockets
`tcp` - TCP protocol
`udp` - User Datagram Protocol for IPv4
`udplite` - Lightweight User Datagram Protocol
`unix` - sockets for local interprocess communication
`vsock` - Linux VSOCK address family
`x25` - ITU-T X.25 / ISO/IEC 8208 protocol interface

`sys/stat.h`

`chmod` - change permissions of a file
`fchmod` - change permissions of a file
`fchmodat` - change permissions of a file
`fstat` - get file status
`fstat64` - get file status
`fstatat` - get file status
`fstatat64` - get file status
`futimens` - change file timestamps with nanosecond precision
`isfdtype` - test file type of a file descriptor
`lstat` - get file status
`lstat64` - get file status
`mkdir` - create a directory
`mkdirat` - create a directory
`mknod` - create a special or ordinary file
`mknodat` - create a special or ordinary file
`newfstatat` - get file status
`oldfstat` - get file status
`oldlstat` - get file status
`oldstat` - get file status
`stat` - get file status
`stat64` - get file status
`statx` - get file status (extended)
`umask` - set file mode creation mask
`utimensat` - change file timestamps with nanosecond precision

sys/statvfs.h

`fstatvfs` - get filesystem statistics

`statvfs` - get filesystem statistics

sys/time.h

`adjtime` - correct the time to synchronize the system clock

`futimes` - change file timestamps

`futimesat` - change timestamps of a file relative to a directory fi...

`getitimer` - get or set value of an interval timer

`gettimeofday` - get / set time

`lutimes` - change file timestamps

`setitimer` - get or set value of an interval timer

`settimeofday` - get / set time

`timeradd` - timeval operations

`timerclear` - timeval operations

`timercmp` - timeval operations

`timerisset` - timeval operations

`timersub` - timeval operations

`TIMESPEC_TO_TIMEVAL` - convert between time structures

`TIMEVAL_TO_TIMESPEC` - convert between time structures

sys/times.h

`times` - get process times

sys/types.h

`bindresvport` - bind a socket to a privileged IP port

`closedir` - close a directory

`db` - database access methods

`dbopen` - database access methods

`dirfd` - get directory stream file descriptor

`endgrent` - get group file entry

`endpwent` - get password file entry

`fdopendir` - open a directory
`freeaddrinfo` - network address and service translation
`freehostent` - get network hostnames and addresses
`freeifaddrs` - get interface addresses
`fts` - traverse a file hierarchy
`fts_children` - traverse a file hierarchy
`fts_close` - traverse a file hierarchy
`fts_open` - traverse a file hierarchy
`fts_read` - traverse a file hierarchy
`fts_set` - traverse a file hierarchy
`gai_strerror` - network address and service translation
`getaddrinfo` - network address and service translation
`getgrent` - get group file entry
`getgrgid` - get group file entry
`getgrgid_r` - get group file entry
`getgrnam` - get group file entry
`getgrnam_r` - get group file entry
`getifaddrs` - get interface addresses
`getipnodebyaddr` - get network hostnames and addresses
`getipnodebyname` - get network hostnames and addresses
`getpw` - reconstruct password line entry
`getpwent` - get password file entry
`getpwnam` - get password file entry
`getpwnam_r` - get password file entry
`getpwuid` - get password file entry
`getpwuid_r` - get password file entry
`initgroups` - initialize the supplementary group access list
`lseek64` - reposition 64-bit read/write file offset
`mkfifo` - make a FIFO special file (a named pipe)
`mkfifoat` - make a FIFO special file (a named pipe)
`opendir` - open a directory
`re_comp` - BSD regex functions
`re_exec` - BSD regex functions
`rewinddir` - reset directory stream
`setgrent` - get group file entry
`setpwent` - get password file entry
`ustat` - get filesystem statistics

sys/uio.h

`preadv` - read or write data into multiple buffers
`preadv2` - read or write data into multiple buffers
`process_vm_readv` - transfer data between process address spaces
`process_vm_writev` - transfer data between process address spaces
`pwritev` - read or write data into multiple buffers
`pwritev2` - read or write data into multiple buffers
`readv` - read or write data into multiple buffers
`writev` - read or write data into multiple buffers

sys/utsname.h

`oldolduname` - get name and information about current kernel
`olduname` - get name and information about current kernel
`uname` - get name and information about current kernel

sys/wait.h

`wait` - wait for process to change state
`wait3` - wait for process to change state, BSD style
`wait4` - wait for process to change state, BSD style
`waitid` - wait for process to change state
`waitpid` - wait for process to change state

syslog.h

`closelog` - send messages to the system logger
`openlog` - send messages to the system logger
`setlogmask` - set log priority mask
`syslog` - send messages to the system logger
`vsylog` - send messages to the system logger

termios.h

`cfgetispeed` - get and set terminal attributes, line control, get and...
`cfgetospeed` - get and set terminal attributes, line control, get and...
`cfmakeraw` - get and set terminal attributes, line control, get and...

`cfsetispeed` - get and set terminal attributes, line control, get and...
`cfsetospeed` - get and set terminal attributes, line control, get and...
`cfsetspeed` - get and set terminal attributes, line control, get and...
`tcdrain` - get and set terminal attributes, line control, get and...
`tcflow` - get and set terminal attributes, line control, get and...
`tcflush` - get and set terminal attributes, line control, get and...
`tcgetattr` - get and set terminal attributes, line control, get and...
`tcgetsid` - get session ID
`tcsendbreak` - get and set terminal attributes, line control, get and...
`tcsetattr` - get and set terminal attributes, line control, get and...
`termios` - get and set terminal attributes, line control, get and...

time.h

`asctime` - transform date and time to broken-down time or ASCII
`asctime_r` - transform date and time to broken-down time or ASCII
`clock` - determine processor time
`clock_getcpuclockid` - obtain ID of a process CPU-time clock
`clock_getres` - clock and time functions
`clock_gettime` - clock and time functions
`clock_nanosleep` - high-resolution sleep with specifiable clock
`clock_settime` - clock and time functions
`ctime` - transform date and time to broken-down time or ASCII
`ctime_r` - transform date and time to broken-down time or ASCII
`daylight` - initialize time conversion information
`difftime` - calculate time difference
`dysize` - get number of days for a given year
`getdate` - convert a date-plus-time string to broken-down time
`getdate_err` - convert a date-plus-time string to broken-down time
`getdate_r` - convert a date-plus-time string to broken-down time
`gmtime` - transform date and time to broken-down time or ASCII
`gmtime_r` - transform date and time to broken-down time or ASCII
`localtime` - transform date and time to broken-down time or ASCII
`localtime_r` - transform date and time to broken-down time or ASCII
`mktime` - transform date and time to broken-down time or ASCII
`nanosleep` - high-resolution sleep
`stime` - set time
`strftime` - format date and time
`strftime_l` - format date and time

`strptime` - convert a string representation of time to a time tm s...
`time` - get time in seconds
`timegm` - inverses of gmtime and localtime
`timelocal` - inverses of gmtime and localtime
`timer_create` - create a POSIX per-process timer
`timer_delete` - delete a POSIX per-process timer
`timer_getoverrun` - get overrun count for a POSIX per-process timer
`timer_gettime` - arm/disarm and fetch state of POSIX per-process timer
`timer_settime` - arm/disarm and fetch state of POSIX per-process timer
`timezone` - initialize time conversion information
`tzname` - initialize time conversion information
`tzset` - initialize time conversion information

ulimit.h

`ulimit` - get and set user limits

unistd.h

`_Exit` - terminate the calling process
`_exit` - terminate the calling process
`_llseek` - reposition read/write file offset
`_sysctl` - read/write system parameters
`access` - check user's permissions for a file
`acct` - switch process accounting on or off
`alarm` - set an alarm clock for delivery of a signal
`arch_prctl` - set architecture-specific thread state
`brk` - change data segment size
`capget` - set/get capabilities of thread(s)
`capset` - set/get capabilities of thread(s)
`chdir` - change working directory
`chown` - change ownership of a file
`chown32` - change ownership of a file
`chroot` - change root directory
`close` - close a file descriptor
`close_range` - close all file descriptors in a given range
`confstr` - get configuration dependent string variables
`copy_file_range` - Copy a range of data from one file to another

`cuserid` - get username
`daemon` - run in the background
`delete_module` - unload a kernel module
`dup` - duplicate a file descriptor
`dup2` - duplicate a file descriptor
`dup3` - duplicate a file descriptor
`eaccess` - check effective user's permissions for a file
`encrypt` - encrypt 64-bit messages
`encrypt_r` - encrypt 64-bit messages
`endusershell` - get permitted user shells
`euidaccess` - check effective user's permissions for a file
`exec` - execute a file
`execl` - execute a file
`execle` - execute a file
`execlp` - execute a file
`execv` - execute a file
`execve` - execute program
`execveat` - execute program relative to a directory file descriptor
`execvp` - execute a file
`execvpe` - execute a file
`exit` - terminate the calling process
`exit_group` - exit all threads in a process
`faccessat` - check user's permissions for a file
`faccessat2` - check user's permissions for a file
`fchdir` - change working directory
`fchown` - change ownership of a file
`fchown32` - change ownership of a file
`fchownat` - change ownership of a file
`fdatasync` - synchronize a file's in-core state with storage device
`fexecve` - execute program specified via file descriptor
`finit_module` - load a kernel module
`fork` - create a child process
`fpathconf` - get configuration values for files
`fsync` - synchronize a file's in-core state with storage device
`ftruncate` - truncate a file to a specified length
`ftruncate64` - truncate a file to a specified length
`futex` - fast user-space locking
`get_current_dir_name` - get current working directory
`get_robust_list` - get/set list of robust futexes
`get_thread_area` - manipulate thread-local storage information

`getcwd` - get current working directory
`getcwd` - get current working directory
`getdents` - get directory entries
`getdents64` - get directory entries
`getdomainname` - get/set NIS domain name
`getdtablesize` - get file descriptor table size
`getegid` - get group identity
`getegid32` - get group identity
`getentropy` - fill a buffer with random bytes
`geteuid` - get user identity
`geteuid32` - get user identity
`getgid` - get group identity
`getgid32` - get group identity
`getgroups` - get/set list of supplementary group IDs
`getgroups32` - get/set list of supplementary group IDs
`gethostid` - get or set the unique identifier of the current host
`gethostname` - get/set hostname
`getlogin` - get username
`getlogin_r` - get username
`getopt` - Parse command-line options
`getopt_long` - Parse command-line options
`getopt_long_only` - Parse command-line options
`getpagesize` - get memory page size
`getpass` - get a password
`getpgid` - set/get process group
`getpgrp` - set/get process group
`getpid` - get process identification
`getppid` - get process identification
`getresgid` - get real, effective, and saved user/group IDs
`getresgid32` - get real, effective, and saved user/group IDs
`getresuid` - get real, effective, and saved user/group IDs
`getresuid32` - get real, effective, and saved user/group IDs
`getsid` - get session ID
`gettid` - get thread identification
`getuid` - get user identity
`getuid32` - get user identity
`getusershell` - get permitted user shells
`getwd` - get current working directory
`group_member` - test whether a process is in a group
`idle` - make process 0 idle

`init_module` - load a kernel module

`io_cancel` - cancel an outstanding asynchronous I/O operation

`io_destroy` - destroy an asynchronous I/O context

`io_getevents` - read asynchronous I/O events from the completion queue

`ioprio_get` - get/set I/O scheduling class and priority

`ioprio_set` - get/set I/O scheduling class and priority

`ipc` - System V IPC system calls

`isatty` - test whether a file descriptor refers to a terminal

`kcmp` - compare two processes to determine if they share a ker...

`kexec_file_load` - load a new kernel for later execution

`kexec_load` - load a new kernel for later execution

`keyctl` - manipulate the kernel's key management facility

`klogctl` - read and/or clear kernel message ring buffer; set cons...

`lchown` - change ownership of a file

`lchown32` - change ownership of a file

`link` - make a new name for a file

`linkat` - make a new name for a file

`llseek` - reposition read/write file offset

`lockf` - apply, test or remove a POSIX lock on an open file

`lookup_dcookie` - return a directory entry's path

`lseek` - reposition read/write file offset

`membarrier` - issue memory barriers on a set of threads

`memfd_secret` - create an anonymous RAM-based file to access secret me...

`mmap2` - map files or devices into memory

`modify_ldt` - get or set a per-process LDT entry

`mount_setattr` - change properties of a mount or mount tree

`mq_getsetattr` - get/set message queue attributes

`nice` - change process priority

`openat2` - open and possibly create a file (extended)

`optarg` - Parse command-line options

`opterr` - Parse command-line options

`optind` - Parse command-line options

`optopt` - Parse command-line options

`pathconf` - get configuration values for files

`pause` - wait for signal

`perf_event_open` - set up performance monitoring

`pidfd_getfd` - obtain a duplicate of another process's file descriptor

`pidfd_open` - obtain a file descriptor that refers to a process

`pidfd_send_signal` - send a signal to a process specified by a file
descri...

`pipe` - create pipe
`pipe2` - create pipe
`pivot_root` - change the root mount
`pread` - read from or write to a file descriptor at a given offset
`pread64` - read from or write to a file descriptor at a given offset
`profil` - execution time profile
`pwrite` - read from or write to a file descriptor at a given offset
`pwrite64` - read from or write to a file descriptor at a given offset
`read` - read from a file descriptor
`readdir` - read directory entry
`readlink` - read value of a symbolic link
`readlinkat` - read value of a symbolic link
`reboot` - reboot or enable/disable Ctrl-Alt-Del
`rmdir` - delete a directory
`rt_sigqueueinfo` - queue a signal and data
`rt_tgsigqueueinfo` - queue a signal and data
`s390_guarded_storage` - operations with z/Architecture guarded storage fac...
`s390_pci_mmio_read` - transfer data to/from PCI MMIO memory page
`s390_pci_mmio_write` - transfer data to/from PCI MMIO memory page
`s390_runtime_instr` - enable/disable s390 CPU run-time instrumentation
`s390_sthyi` - emulate STHYI instruction
`sbrk` - change data segment size
`sched_getattr` - set and get scheduling policy and attributes
`sched_setattr` - set and get scheduling policy and attributes
`seccomp` - operate on Secure Computing state of the process
`set_robust_list` - get/set list of robust futexes
`set_thread_area` - manipulate thread-local storage information
`set_tid_address` - set pointer to thread ID
`setdomainname` - get/set NIS domain name
`setegid` - set effective user or group ID
`seteuid` - set effective user or group ID
`setgid` - set group identity
`setgid32` - set group identity
`setgroups` - get/set list of supplementary group IDs
`setgroups32` - get/set list of supplementary group IDs
`sethostid` - get or set the unique identifier of the current host
`sethostname` - get/set hostname
`setkey` - encrypt 64-bit messages
`setkey_r` - encrypt 64-bit messages

`setpgid` - set/get process group
`setpgrp` - set/get process group
`setregid` - set real and/or effective user or group ID
`setregid32` - set real and/or effective user or group ID
`setresgid` - set real, effective, and saved user or group ID
`setresgid32` - set real, effective, and saved user or group ID
`setresuid` - set real, effective, and saved user or group ID
`setresuid32` - set real, effective, and saved user or group ID
`setreuid` - set real and/or effective user or group ID
`setreuid32` - set real and/or effective user or group ID
`setsid` - creates a session and sets the process group ID
`setuid` - set user identity
`setuid32` - set user identity
`setup` - setup devices and filesystems, mount root filesystem
`setusershell` - get permitted user shells
`sgetmask` - manipulation of signal mask (obsolete)
`sleep` - sleep for a specified number of seconds
`socketcall` - socket system calls
`spu_create` - create a new spu context
`spu_run` - execute an SPU context
`ssetmask` - manipulation of signal mask (obsolete)
`subpage_prot` - define a subpage protection for an address range
`swab` - swap adjacent bytes
`symlink` - make a new name for a file
`symlinkat` - make a new name for a file
`sync` - commit filesystem caches to disk
`syncfs` - commit filesystem caches to disk
`syscall` - indirect system call
`sysconf` - get configuration information at run time
`sysctl` - read/write system parameters
`syslog` - read and/or clear kernel message ring buffer; set cons...
`tcgetpgrp` - get and set terminal foreground process group
`tcsetpgrp` - get and set terminal foreground process group
`tgkill` - send a signal to a thread
`tkill` - send a signal to a thread
`truncate` - truncate a file to a specified length
`truncate64` - truncate a file to a specified length
`ttyname` - return name of a terminal
`ttyname_r` - return name of a terminal
`ualarm` - schedule signal after given number of microseconds

`unlink` - delete a name and possibly the file it refers to
`unlinkat` - delete a name and possibly the file it refers to
`uselib` - load shared library
`userfaultfd` - create a file descriptor for handling page faults in u...
`usleep` - suspend execution for microsecond intervals
`vfork` - create a child process and block parent
`vhangup` - virtually hangup the current terminal
`write` - write to a file descriptor

utime.h

`utime` - change file last access and modification times
`utimes` - change file last access and modification times

utmpx.h

`getutmp` - copy utmp structure to utmpx, and vice versa
`getutmpx` - copy utmp structure to utmpx, and vice versa

wchar.h

`btowc` - convert single byte to wide character
`fgetws` - read a wide-character string from a FILE stream
`fputws` - write a wide-character string to a FILE stream
`fwide` - set and determine the orientation of a FILE stream
`getwchar` - read a wide character from standard input
`mbrlen` - determine number of bytes in next multibyte character
`mbrtowc` - convert a multibyte sequence to a wide character
`mbsinit` - test for initial shift state
`mbsnrtowcs` - convert a multibyte string to a wide-character string
`mbsrtowcs` - convert a multibyte string to a wide-character string ...
`putwchar` - write a wide character to standard output
`ungetwc` - push back a wide character onto a FILE stream
`wcpcpy` - copy a wide-character string, returning a pointer to i...
`wcpncpy` - copy a fixed-size string of wide characters, returning...
`wcrtomb` - convert a wide character to a multibyte sequence
`wcscasecmp` - compare two wide-character strings, ignoring case

`wscat` - concatenate two wide-character strings
`wchr` - search a wide character in a wide-character string
`wscmp` - compare two wide-character strings
`wscpy` - copy a wide-character string
`wscspn` - search a wide-character string for any of a set of wid...
`wsdup` - duplicate a wide-character string
`wslen` - determine the length of a wide-character string
`wscncasecmp` - compare two fixed-size wide-character strings, ignorin...
`wscncat` - concatenate two wide-character strings
`wscncmp` - compare two fixed-size wide-character strings
`wscncpy` - copy a fixed-size string of wide characters
`wcsnlen` - determine the length of a fixed-size wide-character st...
`wcsnrtombs` - convert a wide-character string to a multibyte string
`wcspbrk` - search a wide-character string for any of a set of wid...
`wcsrchr` - search a wide character in a wide-character string
`wcsrtombs` - convert a wide-character string to a multibyte string
`wcsspn` - get length of a prefix wide-character substring
`wcsstr` - locate a substring in a wide-character string
`wcstok` - split wide-character string into tokens
`wcswidth` - determine columns needed for a fixed-size wide-charact...
`wctob` - try to represent a wide character as a single byte
`wcwidth` - determine columns needed for a wide character
`wmemchr` - search a wide character in a wide-character array
`wmemcmp` - compare two arrays of wide-characters
`wmemcpy` - copy an array of wide-characters
`wmemmove` - copy an array of wide-characters
`wmemset` - fill an array of wide-characters with a constant wide ...

wctype.h

`iswalnum` - test for alphanumeric wide character
`iswalpha` - test for alphabetic wide character
`iswblank` - test for whitespace wide character
`iswcntrl` - test for control wide character
`iswctype` - wide-character classification
`iswdigit` - test for decimal digit wide character
`iswgraph` - test for graphic wide character
`iswlower` - test for lowercase wide character
`iswprint` - test for printing wide character

`iswpunct` - test for punctuation or symbolic wide character
`iswspace` - test for whitespace wide character
`iswupper` - test for uppercase wide character
`iswxdigit` - test for hexadecimal digit wide character
`towctrans` - wide-character transliteration
`towlower` - convert a wide character to lowercase
`towlower_l` - convert a wide character to lowercase
`towupper` - convert a wide character to uppercase
`towupper_l` - convert a wide character to uppercase
`wctrans` - wide-character translation mapping
`wctype` - wide-character classification

wordexp.h

`wordexp` - perform word expansion like a posix-shell
`wordfree` - perform word expansion like a posix-shell